Practices (techniques) in agile frameworks

Lecture 8

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Selecting the Right Technique

It can be challenging to select the right technique for the situation to get the desired outcome. The following Table 1 provides general guidance on when to choose a specific technique. Please note that many techniques can be used successfully in a variety of contexts.

An agile business analysis practitioner uses his or her domain knowledge, experience, individual competencies, and creativity when selecting the appropriate technique for the desired outcome. The purpose of Table 1 is to provide a starting point when determining which technique might be useful in a given context.

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Selecting the Right Technique

Context	Works well with internal teams	Works well with stakeholders external to the team
Communications	7.1. Backlog Refinement 7.8. Planning Workshops 7.9. Portfolio Kanban 7.14. Retrospectives 7.24. Visioning	7.15. Reviews 7.24. Visioning
Process Analysis	7.23. Value Stream Mapping	7.3. Impact Mapping
Product Management Refinement	7.6. Minimal Viable Product 7.10. Product Roadmap 7.11. Purpose Alignment Model 7.12. Real Options	7.5. Kano Analysis 7.6. Minimal Viable Product
Requirements Management	7.2. Behaviour Driven Development 7.4. Job Stories 7.13. Relative Estimation 7.16. Spikes 7.18. Story Decomposition 7.19. Story Elaboration 7.20. Story Mapping 7.21. User Stories	
Understanding your Customer	7.7. Personas 7.17. Storyboarding 7.22. Value Modelling	

Table 1. Selecting the Right Technique

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Personas

Purpose

Personas are used to understand and empathize with an intended stakeholder in order to align the solution with the stakeholder need.

Description

Personas are fictional characters or archetypes that exemplify the way that typical users interact with a solution. They are often used in agile approaches to understand value from the perspective of a particular stakeholder and allow a team that may not have direct access to a customer representative to better understand their needs. Work can then focus on the features of greatest value to a particular persona.

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Personas

A persona is described as though it is real person. Personas may provide a name, personality, family, work background, skill level, preferences, behavior patterns, personal attitudes, goals, and needs. They can also include a picture and a short “day in the life” narrative that helps to visualize the user and their experience.

Business analysis practitioners use personas to gain a deeper understanding of stakeholders than is generally provided from a role or actor description. Personas help improve a solution, a purpose, and usability because they are patterned after the subtle qualities of real people that will interact with the solution and how they do their job.

Personas are ranked to identify those that will realize the most benefit from the solution design.

Personas

Elements

1 Long and Short Template

Business analysis practitioners frequently use two different templates when creating personas:

- a short template which offers a one-page quick view of key information, and
- a long template which offers more in-depth details and understanding.

Additional user research helps expand from short template to long template.

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Personas

Name	Details or Context
	• Tasks and responsibilities • Technology
Traits and characteristics	**Goals**
• Age • Occupation • Location	• Goal 1 • Goal 2 • Goal 3

Figure 1. Short Persona Template

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Personas

	Name Demographics	Buying motive	
Traits	**Needs**	**Motivation**	**Differentiators**

Figure 2. Long Persona Template

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Personas

2 Persona Name and Image

Business analysis practitioners give personas a realistic name and attach a fictional image in order increase its relatability and thereby the understanding of and empathy with the intended stakeholder.

3 Traits and Characteristics

Personas include unique, distinguishing, and differentiating characteristics or traits regarding the intended stakeholder.

For example, one ATM persona may be an office manager who deposits cash at the end of the day. A different ATM persona may be an individual who likes to get small amounts out at a time.

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Personas

4 Motivations

Personas include a representation of the underlying motivations regarding how and why the intended stakeholder interacts with the solution.

For example, an ATM office manager may be motivated by reducing risk of fraud or theft. The individual withdrawing from the ATM may be motivated by withdrawing the minimum amount needed.

5 Needs

Needs for the persona address very specific needs. These can be basic needs such as safety, trust, or access to food and shelter. They can be higher level needs such as the need for acceptance and validation. Needs are finite as compared to wants which are infinite.

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Personas

6 Differentiators

Differentiators identify specifically why this persona is different from another persona. They identify what is unique about this persona. These could be generational or experiential differentiators, preference differentiators, or identifying characteristics.

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Personas

Usage Considerations

1 Strengths

• Personas facilitate the shared understanding of specific requirements for different sets of users. These requirements can be used to develop user stories.

• Proposed solutions can be guided by how well they meet the needs of individual user personas. Features can be prioritized based on how well they address the needs of one or more personas.

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Personas

• Provide a human “face” so as to focus empathy on the people represented by the demographics.

• If the data is available, using demographic (or anthropomorphic) data about the intended user population is a good way to start building personas.

However, in some cases it is necessary to be creative and invent personas based on little more than a few dry facts about the intended end users. In either case, a representative pool of personas should be identified.

• Personas help stakeholders from projecting individual values and biases onto the solution. They help to develop compassion for various users.

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Personas

2 Limitations

• Personas are fictional, so there is a tendency to create personas that embody traits common to most users, but this creates a generic user that is not distinct or realistic. This can lead to solutions that are trying to be everything to everyone.

• Personas may not be a good substitute for a real user. Personas can distance a team from a user community.

• Personas need to be regularly reviewed and updated.

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User Stories

Purpose

User Stories are used to convey a customer requirement for the delivery team.

Description

User Stories are a representation of the customer need and are expressed as a small, concise statement of a feature needed to deliver value. User Stories facilitate the interaction and collaboration of stakeholders.

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User Stories

The user story expresses a customer need and value desired. Typically, they are one or more sentences written by the customers, product owners, or business analysis practitioners which describe something of value to a stakeholder. User Stories provide a mechanism for the product owner to scope, coordinate, and prioritize the increments of user value for delivery. A user story is captured on the backlog.

User Stories represent stakeholder needs using short, simple documentation and invite exploration of the requirements through conversations, tests, and supplemental requirements representations as needed. They are concise and easy to change as stakeholder needs are better understood or as those needs evolve.

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User Stories

A commonly used construct for ensuring quality in user stories is the INVEST criteria, which calls for user stories to be:

- (I) Independent: represents a feature which can be delivered independent of other features. Example: "ATM PIN entry" is independent from 'Withdrawal Amount'
- (N) Negotiable: the team can negotiate how to deliver.
- (V) Valuable: expresses the value to the customer. Example: "ATM PIN entry" allows only the correct person to access the account.
- (E) Estimable: team can estimate effort to deliver based on past experience.
- (S) Sized Appropriately: for the team to complete in one iteration. In general, the smaller the better.
- (T) Testable: can be validated objectively by a stakeholder.

Not all backlog items are to be written as user stories. However, User Stories is a common technique used as they emphasize the customer value.

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User Stories

Elements

User Stories follow the 3Cs:

- Card,
- Conversation, and
- Confirmation.

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User Stories

1 Card

Title (optional)

The title of the user story describes an activity that the user wants to carry out with the solution. Typically, it is an active-verb goal phrase, similar to the way use cases are titled.

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User Stories

2 Format

There is no mandatory structure for user stories; however, the most popular format includes three components:

- a user role or persona [WHO],
- a necessary action, behavior, or feature [WHAT], and
- the benefit or business value received by the user when the story is implemented [WHY].

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User Stories

Example format 1

"As a <role>, I need to <feature> so that <goal or value>.“

As a current bank account holder, I need to access my account, so that I can withdraw cash.

Example format 2

"In order to <business value>, as a <role>, I need to <behavior>.“

In order to withdraw cash from my account, as a current bank account holder, I need access to my account.

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User Stories

2 Conversation

The intended purpose of User Stories is to communicate between stakeholders and the delivery team. The user story intentionally does not capture all there is to know about the customer need. A well-written user story invokes conversation among the team.

User Stories

3 Confirmation

Business analysis practitioners confirm the delivered item satisfies the need expressed in the user story. This is most often expressed as acceptance criteria.

Acceptance criteria define the boundaries of a user story and help verify and validate the solution met the intended user need.

Acceptance criteria help the delivery team identify the minimum amount of function necessary. Acceptance criteria are primarily used by the product owner and stakeholders to verify and validate. They also serve as a basis for acceptance tests, regression tests, exploratory tests, and other tests to be developed in support of the product.

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User Stories

5 User Story Management

There are a variety of ways in which user stories are used throughout an agile initiative. The follow techniques are all impacted by User Stories:

- Backlog Refinement
- Product Roadmap
- Story Decomposition,
- Story Elaboration, and
- Visioning

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User Stories

Usage Considerations

1 Strengths

• Tied to small, implementable, and testable slices of functionality facilitating rapid delivery and frequent customer feedback.

• Easily understandable by stakeholders.

• Can be developed through a variety of elicitation techniques, including but not limited to facilitated workshops, contextual inquiry, and other ethnographic elicitation techniques.

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User Stories

• User Stories are simple enough that people can learn to write them in a few minutes, being careful about always delivering business value.

• The process of collaborating on defining and exploring stories builds team commitment and shared understanding of the business domain.

• User Stories invite conversation for further decomposition and exploration.

• To facilitate estimating, planning, and delivery, many agile teams supplement User Stories with analysis models (such as a data model, business rules, user acceptance tests, screen mock-ups or prototypes, context diagram, and state diagram).

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User Stories

Limitations

• This conversational approach can challenge the team, since they do not have all the answers and detailed specifications upfront.

• Too many stories can inflate the backlog.

• User Stories spawn more User Stories through decomposition, so the information must be organized to ensure it is current and relevant.

• The collection of User Stories needs to be regularly refined and prioritized.

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User Stories

• Teams can get too focused on individual User Stories and lose sight of the vision and product roadmap.

• This technique includes the desired feature. Teams can form habits overlooking the value statement (WHY) and instead focus on the feature (WHAT).

References

Agile Extension to the BABOK® Guide (2017). International Institute of Business Analysis and Agile Alliance.